

Dispersive optical model analysis of nucleon scattering on ^{90}Zr

Xiuniao Zhao¹, Wenqing Du¹, R. Capote^{2,*} and E. Sh. Soukhovitskii³

¹College of Physical Science and Technology, Bohai University, Jinzhou 121013, China

²NAPC-Nuclear Data Section, International Atomic Energy Agency, Vienna A-1400, Austria

³Joint Institute for Energy and Nuclear Research, Minsk-Sosny 220109, Belarus



(Received 14 March 2023; accepted 30 May 2023; published 15 June 2023)

^{90}Zr is a double-magic spherical nucleus. Nucleon scattering on such a nucleus can be well described by spherical optical model potentials. Nucleon scattering data including neutron total cross section, elastic scattering angular distributions and analyzing powers, and low-energy scattering observables are used to derive a new Lane-consistent spherical optical model potential that obeys dispersive constraints, features nonlocality of the real part, and considers the shell gap in the potential. It is shown that the derived dispersive optical potential gives a good description of the energies of the bound single-particle states of the target nucleus ^{90}Zr .

DOI: [10.1103/PhysRevC.107.064606](https://doi.org/10.1103/PhysRevC.107.064606)

I. INTRODUCTION

Zirconium has high strength, hardness, ductility, irradiation stability, and melting point, and is resistant to high temperature and water vapor corrosion. Additionally, zirconium features low neutron capture cross sections. Based on the above physical and chemical properties, zirconium is often used as cladding material and core structural material for water-cooled power reactors. Zirconium can also be used as aerospace material, and in innovative reactor designs.

As the most abundant nuclide among the five stable natural isotopes of zirconium, ^{90}Zr is an important nuclide in the structural materials of nuclear power plants, and its nuclear reaction scattering data has important application value. In the past, experimental measurements and theoretical model studies have accumulated a large amount of data. However, significant discrepancies remain between evaluated and measured data that require further improvement of evaluated neutron induced cross sections. An important first step for a new evaluation is to derive an accurate optical model potential capable of describing all available data from negative energies up to 150 MeV of nucleon incident energy.

The phenomenological optical model potential has been successfully used to analyze and calculate nucleon scattering data [1]. Due to the requirement of causality, the dispersion effects need to be considered in nucleon scattering. The dispersive optical model (DOM) potential, which introduces the dispersion relation linking the imaginary potentials to the corresponding real dispersive contributions, has been widely used to describe the nucleon scattering [2–6]. The introduction of the dispersion relation allows considering causality, but also establishes the connection between nuclear reaction and nuclear structure, which makes it possible to simultaneously describe the nucleon scattering and the

single-particle bound states induced by the same nuclear mean field.

Since Mahaux and Sartor proposed that the mean field can be extrapolated from positive towards negative nucleon energies by using dispersion relation, many researchers have tried to establish a unified description of scattering and bound states [7–9]. Among these works we highlight the dispersive optical potentials obtained by Morillon *et al.* [2,10,11] for both neutrons (RIPL 2407 [10] and 2411 [11]) and protons (RIPL 5411 [11]). However, those authors acknowledge the difficulty of deriving a Lane consistent potential.

Some authors developed a Lane-consistent dispersive optical model potential, and obtained a good description of scattering data and bound-state data for nucleon-induced reactions on the double-magic target ^{208}Pb . In particular, the real part of the dispersive optical potential is used as the shell model potential to describe the bound single-particle neutron states in ^{208}Pb [12]. In order to improve the description of the nucleon scattering and the bound-state energies using a dispersive potential, we considered the nonlocality of the real potential and the shell gap in the definition of the imaginary potentials. A newly derived potential for ^{208}Pb gives an even better description of nucleon scattering and the bound-state data [13].

Following our previous work [13], the aim of the present work is to derive a dispersive optical model potential for ^{90}Zr that improves the description of both scattering and bound-state data. We will also test our calculation results by comparing with those using the potentials compiled in the RIPL database, including the Morillon *et al.* [2,10,11] RIPL 2407 and 2411 neutron potentials and RIPL 5411 proton potential. Additionally, the cross sections calculated with the coupled-channels Lane consistent potential proposed by Soukhovitskii *et al.* (RIPL 609/4609 [5]) based on a soft-rotor structure [14,15] will be also considered in the comparison. The Soukhovitskii *et al.* potential RIPL 609 was used in the

*Corresponding author: r.capotenoy@iaea.org

ENDF/B-VIII.0 evaluation of neutron induced reactions on zirconium isotopes [16].

The paper is structured as follows. Section II provides a brief description of the DOM potential. Results and discussions are presented in Sec. III. Finally, Sec. IV contains summary and conclusions.

II. DISPERSIVE OPTICAL MODEL

The spherical dispersive optical model potential used in this work is as follows:

$$\begin{aligned}
 V(r, R, E) = & -V_{\text{HF}}(E^*) \times f_{\text{WS}}(r, R_{\text{HF}}) \\
 & -[\Delta V_v(E^*) + iW_v(E^*)] \times f_{\text{WS}}(r, R_v) \\
 & -[\Delta V_s(E^*) + iW_s(E^*)] \times g_{\text{WS}}(r, R_s) \\
 & + \left(\frac{\hbar}{m_\pi c}\right)^2 [V_{\text{so}}(E^*) + \Delta V_{\text{so}}(E^*) + iW_{\text{so}}(E^*)] \\
 & \times \frac{1}{r} \frac{d}{dr} f_{\text{WS}}(r, R_{\text{so}}) \times (\vec{\sigma} \cdot \vec{L}) + V_{\text{Coul}}(r, R_C).
 \end{aligned} \quad (1)$$

The incident nucleon energy $E^* = E_n$ is equal to the incident energy for neutrons; for incident protons we have to take into account the modification of the nuclear potential by Coulomb repulsion. We assume that the “effective” interacting energy of the proton is $E^* = E_p - C_{\text{Coul}} \frac{Z'}{A^{1/3}}$, where A, Z' are the target mass and atomic numbers. The term $C_{\text{Coul}} \frac{Z'}{A^{1/3}}$ is an estimate of the kinetic energy loss of the incident proton in the interaction region due to Coulomb repulsion. The use of the “effective” interacting energy of the proton is a generalization of the previously used Coulomb corrections, which considers such corrections in all orders. The constant C_{Coul} is an adjustable constant meant to account for the “effective” radius of interaction of proton in nucleus. Avoiding the use of Coulomb corrections is a precondition to the Lane consistency, otherwise the resulting potential is not symmetric with respect to the nucleon charge.

The present optical potential includes relativistic corrections as discussed by Elton [17]. First, the nucleon wave number k was calculated in the relativistic form $(\hbar k)^2 = [E^2 - (M_p c^2)^2]/c^2$, where E denotes the total energy of projectile, M_p the projectile rest mass, and c the light velocity. Second, projectile and target masses were replaced by corresponding relativistic energies in reduced mass formulae. However, the change of the potential depth related to the transformation from the Dirac equation was not considered in this work.

The geometrical Woods-Saxon form factors of the optical model potential are given as

$$f_{\text{WS}}(r, R_i) = \{1 + \exp[(r - R_i)/a_i]\}^{-1}, \quad i = \text{HF}, v, \text{ so} \quad (2)$$

$$g_{\text{WS}}(r, R_s) = -4a_s \frac{d}{dr} f_{\text{WS}}(r, R_s). \quad (3)$$

Nuclear radii $R_i = r_i A^{1/3}$ ($i = \text{HF}, v, s, C, \text{ so}$) are defined for five potential geometries described above, where A is the target mass number. This DOM potential includes the real

$V_{\text{HF}}(E)$ (so called “Hartree-Fock” potential) and imaginary volume (v), surface (s), and spin-orbit (so) potentials, and the corresponding real dispersive contributions $\Delta V_v(E)$ (volume), $\Delta V_s(E)$ (surface), and $\Delta V_{\text{so}}(E)$ (spin-orbit) which are calculated analytically from the corresponding imaginary potentials [18,19]. The general formulation of the Lane-consistent DOM potential has been published before, and more detailed descriptions can be found in Refs. [6,12,20]. Particularly, Zhao *et al.* work considered the nonlocality of the real potential as suggested by Perey and Buck [21]; the local energy approximation of Gaussian type nonlocal potential then results in the following implicit equation to derive the equivalent local V_{HF} potential:

$$V_{\text{HF}}(E) = A_{\text{HF}} \exp\left(-\frac{\mu\beta^2}{(\hbar c)^2} [E - V_{\text{HF}}(E)]\right), \quad (4)$$

where the parameter β is a nonlocality range parameter given in fermi, and A_{HF} is the isovector real constant [(see Eq. (8))]. The shell gap in the definition of the imaginary potentials (and therefore in the corresponding dispersive contribution) near the Fermi energy [22–24] is considered as follows:

$$W_v(E) = \begin{cases} 0, & E_F < E < E_P, \\ A_v \frac{(E-E_P)^2}{(E-E_P)^2 + (B_v)^2}, & E > E_P, \end{cases} \quad (5)$$

$$W_s(E) = \begin{cases} 0, & E_F < E < E_P, \\ A_s \frac{(E-E_P)^2}{(E-E_P)^2 + (B_s)^2} \times \exp(-C_s |E - E_P|), & E > E_P, \end{cases} \quad (6)$$

$$W_{\text{so}}(E) = \begin{cases} 0, & E_F < E < E_P, \\ W_{\text{SO}} \frac{(E-E_P)^2}{(E-E_P)^2 + (B_{\text{so}})^2}, & E > E_P, \end{cases} \quad (7)$$

where $A_v, B_v, B_s, C_s, W_{\text{SO}}$, and B_{so} are undetermined constants, A_s is the isovector surface constant [see Eq. (9)]. The imaginary part of the DOM potential is assumed to be zero in region from $2E_F - E_P$ to E_P around the Fermi energy, and the shell gap $\Delta = 2(E_P - E_F)$. This assumption reflects the physical fact that no excited levels are available for absorption within the magic shell gap. The imaginary part of the DOM potential below the Fermi energy is defined by the symmetry condition $W(2E_F - E) = W(E)$. Following Mahaux and Sartor [23,24], the imaginary volume potential $W_v(E)$ is modified above some fixed energy E_a .

The isospin dependence of the potential (the Lane term [25,26]) was considered in the real potential $V_{\text{HF}}(E)$ and the imaginary surface potential $W_s(E)$ as follows:

$$A_{\text{HF}} = V_0 \left[1 + (-1)^{(Z'+1)} \frac{C_{\text{viso}}}{9V_0} \right], \quad (8)$$

$$A_s = W_0 \left[1 + (-1)^{(Z'+1)} \frac{C_{\text{wiso}}}{9W_0} \right], \quad (9)$$

where $V_0, C_{\text{viso}}, W_0$, are undetermined constants, and it was considered that $(N - Z)/A = 1/9$ for ^{90}Zr .

Optical-model code OPTMAN [27–29] was used for OMP parameter fitting and scattering data calculations. The parameters of the dispersive optical model potential were searched to

TABLE I. Dispersive optical-model potential parameters for nucleon induced reactions on ^{90}Zr .

	Volume	Surface	Spin-orbit	Coulomb
Real potential	$V_0 = 80.43 \text{ MeV}$ $\beta = 0.88 \text{ fm}$ $C_{\text{viso}} = 24 \text{ MeV}$ + dispersive (ΔV_v)	Dispersive (ΔV_s)	$V_{\text{so}} = 7.59 \text{ MeV}$ $\lambda_{\text{so}} = 0.005 \text{ MeV}^{-1}$ + dispersive (ΔV_{so})	$C_{\text{Coul}} = 1.12 \text{ MeV}$
Imaginary potential	$A_v = 9.45 \text{ MeV}$ $B_v = 75.27 \text{ MeV}$ $E_a = 56 \text{ MeV}$ $\alpha = 0.237 \text{ MeV}^{1/2}$	$W_0 = 12.04 \text{ MeV}$ $B_s = 8.59 \text{ MeV}$ $C_s = 0.015 \text{ MeV}^{-1}$ $C_{\text{wiso}} = 22 \text{ MeV}$	$W_{\text{SO}} = -3.1 \text{ MeV}$ $B_{\text{so}} = 160 \text{ MeV}$	
Potential geometry	$r_{\text{HF}} = 1.236 \text{ fm}$ $a_{\text{HF}} = 0.65 \text{ fm}$ $r_v = 1.35 \text{ fm}$ $a_v = 0.5 \text{ fm}$	$r_s = 1.20 \text{ fm}$ $a_s = 0.565 \text{ fm}$	$r_{\text{so}} = 1.11 \text{ fm}$ $a_{\text{so}} = 0.57 \text{ fm}$	$r_C = 1.13 \text{ fm}$ $a_C = 0.44 \text{ fm}$

fit the experimental proton and neutron scattering data by minimizing the quantity χ^2 in a usual way [30]. All experimental data used in the fitting process are taken from the EXFOR database [31]. The newly derived DOM potential (DOMP) parameters are listed in Table I.

Figure 1 shows the energy dependence of DOMP depths and dispersive contributions for neutron induced reactions on ^{90}Zr , and compared with the RIPL 609 potential as well as RIPL 2407 and 2411 potentials. Note that E_P and E_F are different for neutron- and proton-induced reactions, and Table II lists the corresponding parameter values.

III. RESULTS AND DISCUSSION

Figure 2 shows the calculated neutron total cross sections for ^{90}Zr . It is seen that the calculations are in excellent agreement with experimental data from 0.1 to 200 MeV. Similar agreement is also shown for the Soukhovitskii *et al.* (RIPL 609) and Morillon *et al.* (RIPL 2407 and RIPL 2411)

optical model potentials retrieved from the RIPL database. As the incident energy of neutrons decreases, the experimental data show a clear resonance structure. The optical model is not capable of describing observed fluctuating cross sections, but can describe only the average cross sections. Obtained description of the “average” cross section is good.

Figures 3 and 4 shows calculated elastic scattering angular distributions of neutrons and protons incident on ^{90}Zr respectively. Calculations for both neutron and proton elastic scattering describe the experimental data rather well over the entire energy and angular range, with only slight underestimation at low energies less than 5 MeV of neutron incident energy, probably due to missing compound-elastic contribution. Compared with the results calculated by using the RIPL 2407 and 5411 potentials, the current description of angular distribution data is better at large angles. For the description of proton scattering angular distributions, with the increase of incident energy, our description shows slightly higher quality than the coupled-channels RIPL 4609 potential [5].

The imaginary part of the RIPL potentials involved in the comparison lacks consideration of the shell gap of the imaginary part potential, so that the absorption potential near the Fermi energy is overestimated. Therefore, the compensation for the incident channel (i.e., dispersive correction terms) obtained by the dispersion relation is overestimated. Due to the neglect of the shell gap of the V_{HF} part of the RIPL 609 potential, the V_{HF} potential depth of the RIPL 609 potential at positive energies is lower than our present DOMP when fitting the neutron experimental data. Furthermore, the V_{HF} part of the RIPL 2407 and 2411 potentials gives a larger nonlocality range, which makes the V_{HF} potential depth decrease more slowly with the incident energy increase, and is significantly higher than that of our present DOMP when incident energy

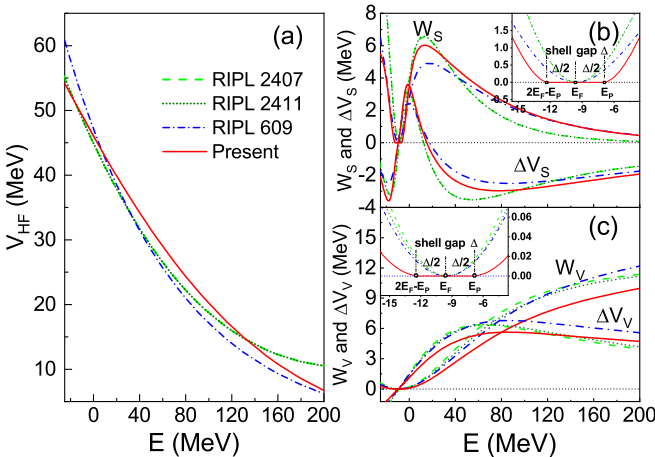


FIG. 1. Energy dependence of Hartree-Fock potential depth, the imaginary potential depths and corresponding dispersive contributions calculated for neutron induced reactions on ^{90}Zr . The smaller plot shows the imaginary potential depths near the Fermi energy E_F .

TABLE II. The parameters E_P and E_F of dispersive optical-model imaginary potential for nucleon induced reactions on ^{90}Zr .

	E_F	E_P
neutron	-9.583 MeV	-7.103 MeV
proton	-6.758 MeV	-5.158 MeV

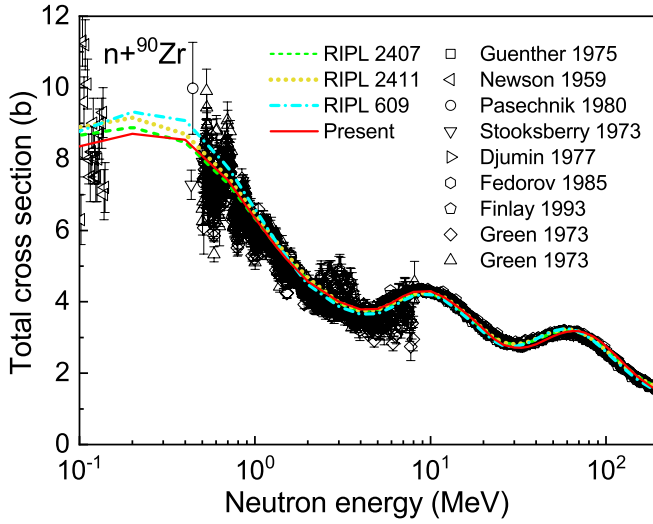


FIG. 2. Comparison of the calculated total cross section for ^{90}Zr with calculations using the RIPL 609 [5], 2407 [10], and 2411 [11] potentials and measurements.

is above about 120 MeV. Consequently, compared to our results, the descriptions of elastic scattering angular distributions obtained by using the RIPL potentials are generally underestimated slightly, except that those calculated by using the RIPL 2407 and 2411 potentials are overestimated at energies above 120 MeV.

The proton elastic scattering analyzing powers for ^{90}Zr are shown in Fig. 5. Current results are in excellent agreement with the experimental data. Our potential achieves a higher quality description of analyzing powers data than the Morillon *et al.* and Soukhovitskii *et al.* potentials, and similar behavior is observed to the one discussed above for angular distributions.

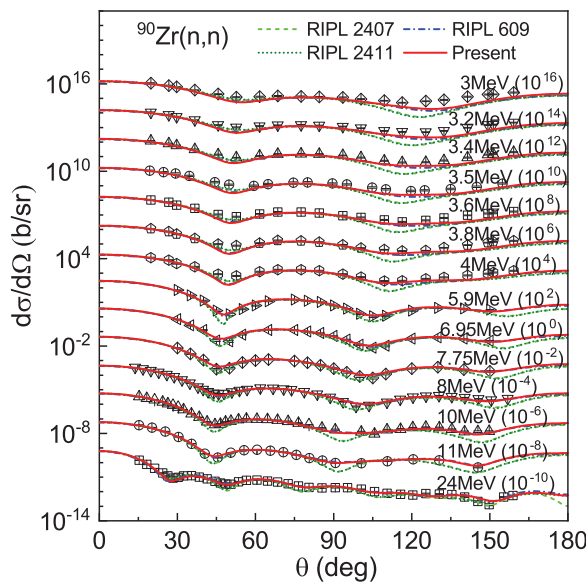


FIG. 3. Comparison of neutron elastic scattering angular distributions for ^{90}Zr with calculations using the RIPL 609 [5], 2407 [10], and 2411 [11] potentials and measurements.

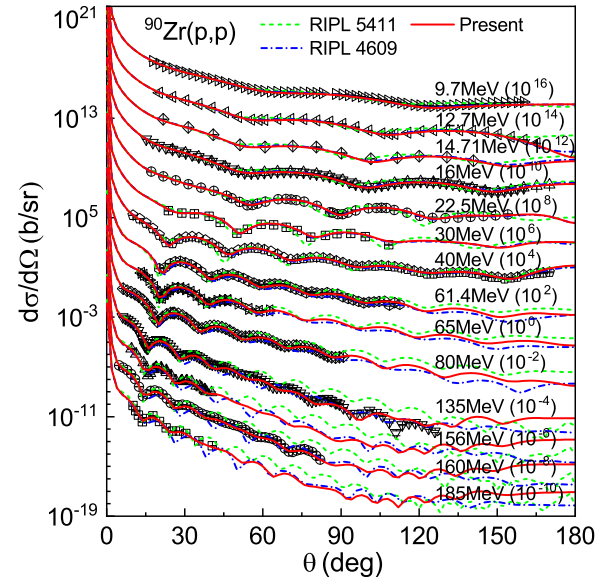


FIG. 4. Comparison of proton elastic scattering angular distributions for ^{90}Zr with calculations using the RIPL 4609 [5] and 5411 [11] potentials and measurements.

Figure 6 show the calculated quasielastic (p,n) angular distribution for scattering on ^{90}Zr , and reasonable agreement with data is achieved which demonstrates the Lane consistency of derived optical model potential. To further improve the description of the quasielastic data, we need to introduce a shift of the isovector and isoscalar geometries; further research on this subject is planned.

Compared with “traditional” dispersive optical potential, the introduction of the shell gap and consideration of the nonlocality of DOM potential improves the description of

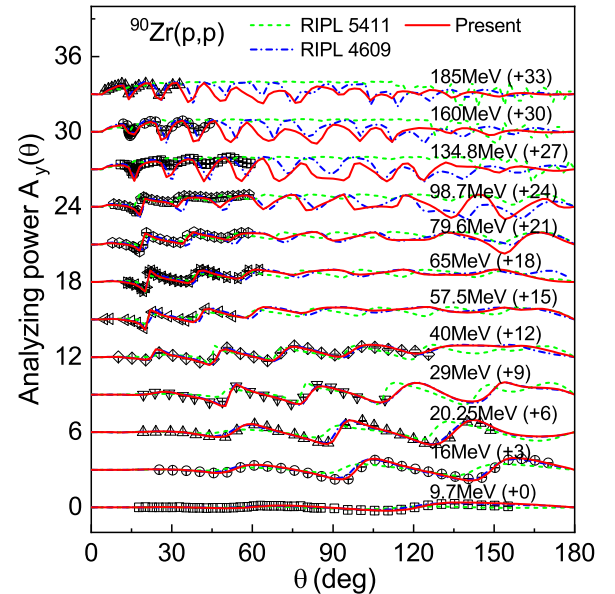


FIG. 5. Proton elastic scattering analyzing powers for ^{90}Zr compared with calculations using the RIPL 4609 [5] and 5411 [11] potentials and the experimental data.

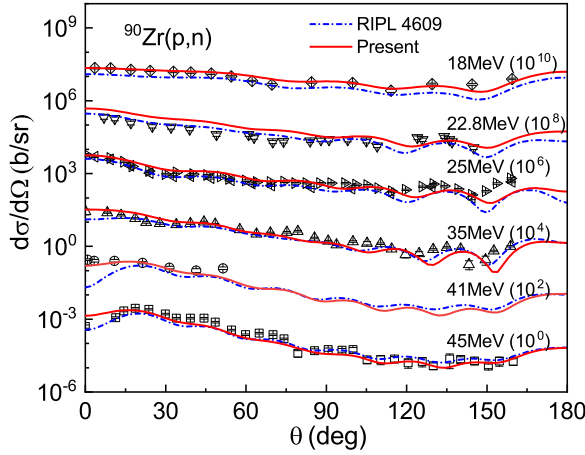


FIG. 6. Angular distributions of the quasielastic (p, n) scattering on ^{90}Zr compared with calculations using the RIPL 4609 [5] potentials and the experimental data.

the scattering cross sections. At the same time, the resulting potential achieve a reasonable description of the bound single-particle states without additional modification. We extracted the real part of the dispersive optical potential derived in this work as the shell model potential, and used this shell model potential to calculate the bound single-particle neutron states for a given spin and parity. This potential includes the sum of the real $V_{\text{HF}}(E_{nlj})$ and spin-orbit $V_{so}(E_{nlj})$ potentials, and all dispersive correction terms $\Delta V_v(E_{nlj})$, $\Delta V_s(E_{nlj})$, and $\Delta V_{so}(E_{nlj})$. Figure 7 shows the comparison of the predicted single-particle (hole) energies with the experimental data and the results obtained using potentials from Morillon *et al.* with labels “M.R.(2006)” [10], “M.R.(2006) (r.s.o + 35%)” [10], and “M.R.(2007)” [11]. The potential with label “M.R.(2006)” is equivalent to the RIPL 2407 potential. However, the calculation result of the third column “(r.s.o + 35%)” in Fig. 7 is obtained with an artificial correction in order to improve the description of single-particle levels, that is, the spin-orbit potential is increased by 35% when calculating the bound single-particle states. And the

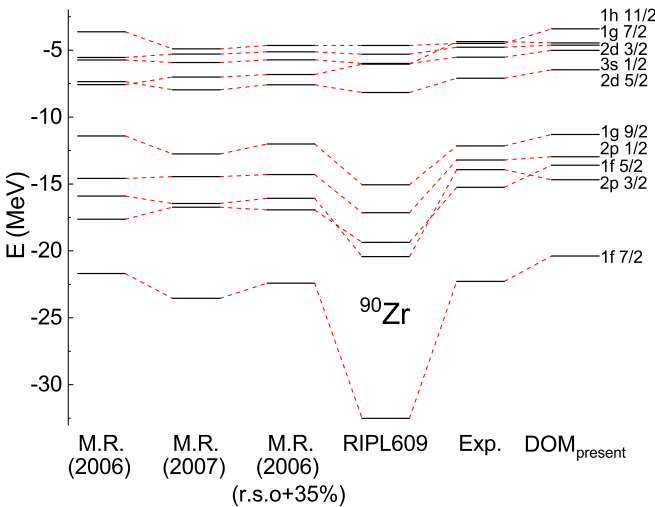


FIG. 7. Neutron single-particle (hole) energies in ^{90}Zr .

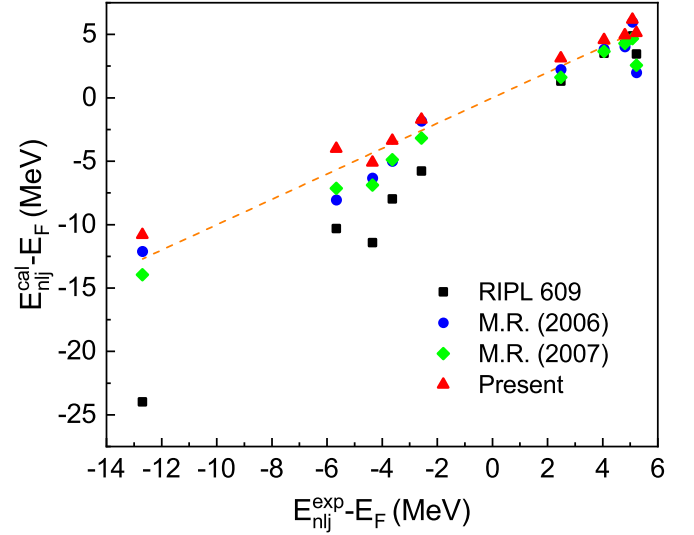


FIG. 8. The dependence upon $E_{nlj}^{\text{exp}} - E_F$ of the difference $E_{nlj}^{\text{cal}} - E_F$. The dashed line represent the average behavior of $E_{nlj}^{\text{cal}} - E_F$ when $E_{nlj}^{\text{cal}} = E_{nlj}^{\text{exp}}$.

potential with label “M.R.(2007)” is similar to the RIPL 2411 potential, RIPL 2411 potential slightly modifies the parameters of imaginary part potential in order to try to build a Lane-consistent optical potential. These approaches make the real part of the optical potential for the scattering state not equivalent to the shell model potential for the bound state. In particular, considering that the deformation of the double-magic nucleus is small, we ignore its dynamic deformation and approximately regard RIPL 609 potential as spherical. The calculated single-particle energies when the real part of the RIPL 609 potential is used as the shell model potential are also shown in Fig. 7.

It can be seen from Fig. 7 that derived bound-state data are in reasonable agreement with the experimental data; the results are similar to those of the Morillon *et al.* potentials and significantly better than those of the RIPL 609 potential. The nonlocality of the real V_{HF} part and the shell gap were not considered in the RIPL 609 potential, which makes the coupling between single-particle (hole) energy levels (essentially nucleon-nucleon interaction) overestimated. As a result the real potential depth becomes too deep, leading to calculated single-hole energies deeper than the experimental value. Although our results for some single-particle levels still exhibit an inversion of the experimentally known sequence, the overall description of the bound single-particle states’ levels is improved. The order inversion of the levels $1f_{5/2}$ and $2p_{3/2}$ is similar to that calculated by Delaroche *et al.* [32]. The difference between the single-particle (hole) energies E_{nlj} and the Fermi energy E_F is defined. Figure 8 shows the dependence upon experimental energies $E_{nlj}^{\text{exp}} - E_F$ of the difference calculated energies $E_{nlj}^{\text{cal}} - E_F$, which further shows that results from the current work are in good agreement with the experimental data.

The bound single-particle state data were not considered during the parameters adjustment. We pursued the optimal description of the scattering state data, and used the real part

of the obtained optical potential directly as the shell model potential for the calculation of the single-particle bound neutron states. Readjusting parameters by considering bound-state data can improve the description for single-particle bound states, but at the same time readjustment will sacrifice the quality of the description for scattering states to a certain extent. Further improvement is warranted.

IV. CONCLUSIONS

Following our previous work [13], a dispersive optical model potential is used to describe the nucleon scattering on the double-magic nucleus ^{90}Zr , which considers the nonlocality in the real V_{HF} potential and introduces the shell gap in the definition of the nuclear imaginary potentials near the Fermi energy. The newly derived potential shows a very good description of available nucleon-nucleus elastic scattering

data up to 200 MeV, including neutron total cross sections, nucleon elastic scattering angular distributions, and analyzing powers. Lane consistency of the derived potential is demonstrated by a good agreement of calculated quasielastic (p, n) angular distributions of the isobaric analog state for different proton incident energies. In addition, the real part of the DOM potential is used as the shell model potential to calculate the single-particle (hole) energies, and gives a reasonably good description of available experimental data. Further work is warranted to extend the derived potential to other zirconium isotopes.

ACKNOWLEDGMENT

This work is the final result of extended and fruitful collaboration over many years with our colleague Efrem Sh. Soukhovitskiĭ who recently passed away.

-
- [1] A. J. Koning and J. P. Delaroche, Local and global nucleon optical models from 1 keV to 200 MeV, *Nucl. Phys. A* **713**, 231 (2003).
 - [2] B. Morillon and P. Romain, Dispersive and global spherical optical model with a local energy approximation for the scattering of neutrons by nuclei from 1 keV to 200 MeV, *Phys. Rev. C* **70**, 014601 (2004).
 - [3] E. Sh. Soukhovitskiĭ, R. Capote, J. M. Quesada, and S. Chiba, Dispersive coupled-channel analysis of nucleon scattering from ^{232}Th up to 200 MeV, *Phys. Rev. C* **72**, 024604 (2005).
 - [4] R. Capote, E. Sh. Soukhovitskiĭ, J. M. Quesada, and S. Chiba, Is a global coupled-channel dispersive optical model potential for actinides feasible?, *Phys. Rev. C* **72**, 064610 (2005).
 - [5] R. Capote *et al.*, RIPL–Reference Input Parameter Library for calculation of nuclear reactions and nuclear data evaluations, *Nucl. Data Sheets* **110**, 3107 (2009).
 - [6] E. Sh. Soukhovitskiĭ, R. Capote, J. M. Quesada, S. Chiba, and D. S. Martyanov, Nucleon scattering on actinides using a dispersive optical model with extended couplings, *Phys. Rev. C* **94**, 064605 (2016).
 - [7] C. Mahaux and R. Sartor, Calculation of the Shell-Model Potential from the Optical-Model Potential, *Phys. Rev. Lett.* **57**, 3015 (1986).
 - [8] M. H. Mahzoon, R. J. Charity, W. H. Dickhoff, H. Dussan, and S. J. Waldecker, Forging the Link between Nuclear Reactions and Nuclear Structure, *Phys. Rev. Lett.* **112**, 162503 (2014).
 - [9] W. H. Dickhoff and R. J. Charity, Recent developments for the optical model of nuclei, *Prog. Part. Nucl. Phys.* **105**, 252 (2019).
 - [10] B. Morillon and P. Romain, Bound single-particle states for neutrons from a global spherical optical model, *Phys. Rev. C* **74**, 014601 (2006).
 - [11] B. Morillon and P. Romain, Bound single-particle states and scattering of nucleons on spherical nuclei with a global optical model, *Phys. Rev. C* **76**, 044601 (2007).
 - [12] X. Zhao, W. Sun, E. Sh. Soukhovitskiĭ, D. S. Martyanov, J. M. Quesada, and R. Capote, Analysis of neutron bound states of ^{208}Pb by a dispersive optical model potential, *J. Phys. G: Nucl. Part. Phys.* **46**, 055103 (2019).
 - [13] X. Zhao, W. Sun, R. Capote, E. Sh. Soukhovitskiĭ, D. S. Martyanov, and J. M. Quesada, Dispersive optical model description of nucleon scattering on Pb and Bi isotopes, *Phys. Rev. C* **101**, 064618 (2020).
 - [14] Yu. V. Porodzinskii and E. Sh. Soukhovitskiĭ, Rotational-vibrational states of nonaxial deformable even-even nuclei, *Sov. J. Nucl. Phys.* **53**, 41 (1991).
 - [15] Yu. V. Porodzinskij and E. Sh. Soukhovitskiĭ, Analysis of neutron scattering by even-even nuclei with allowance for dynamical octupole deformations, *Phys. At. Nucl.* **59**, 228 (1996).
 - [16] D. A. Brown, M. B. Chadwick, R. Capote *et al.*, ENDF/B-VIII.0: The 8th major release of the nuclear reaction data library with CIELO-project cross sections, new standards and thermal scattering data, *Nucl. Data Sheets* **148**, 1 (2018).
 - [17] L. R. B. Elton, The optical model at relativistic energies, *Nuovo Cimento B* **43**, 277 (1966).
 - [18] J. M. Quesada, R. Capote, A. Molina, M. Lozano, and J. Raynal, Analytical expressions for the dispersive contributions to the nucleon-nucleus optical potential, *Phys. Rev. C* **67**, 067601 (2003).
 - [19] J. M. Quesada, R. Capote, A. Molina, and M. Lozano, Dispersion relations in the nuclear optical model, *Comput. Phys. Commun.* **153**, 97 (2003).
 - [20] E. Sh. Soukhovitskiĭ, R. Capote, J. M. Quesada, S. Chiba, and D. S. Martyanov, Erratum: Nucleon scattering on actinides using a dispersive optical model with extended couplings [*Phys. Rev. C* **94**, 064605 (2016)], *Phys. Rev. C* **102**, 059901 (2020).
 - [21] F. Perey and B. Buck, A non-local potential model for the scattering of neutrons by nuclei, *Nucl. Phys.* **32**, 353 (1962).
 - [22] A. Molina, R. Capote, J. M. Quesada, and M. Lozano, Dispersive spherical optical model of neutron scattering from ^{27}Al up to 250 MeV, *Phys. Rev. C* **65**, 034616 (2002).
 - [23] C. Mahaux and R. Sartor, Dispersion relation approach to the mean field and spectral functions of nucleons in ^{40}Ca , *Nucl. Phys. A* **528**, 253 (1991).
 - [24] C. Mahaux and R. Sartor, *Advances in Nuclear Physics*, edited by J. W. Negele and E. Vogt, Vol. 20 (Plenum, New York, 1991).
 - [25] A. M. Lane, New Term in the Nuclear Optical Potential: Implications for (p, n) Mirror State Reactions, *Phys. Rev. Lett.* **8**, 171 (1962).
 - [26] A. M. Lane, Isobaric spin dependence of the optical potential and quasi-elastic (p, n) reactions, *Nucl. Phys.* **35**, 676 (1962).

- [27] E. Sh. Soukhovitskiĭ, G. B. Morogovskii, S. Chiba, O. Iwamoto, and T. Fukahori, Physics and Numerical Methods of OPTMAN: A Coupled-channels Method Based on Soft-rotator Model for a Description of Collective Nuclear Structure and Excitation, Japan Atomic Energy Research Institute Technical Report No. JAERI-Data/Code 2004-002, 2004, <https://jopss.jaea.go.jp/pdfdata/JAERI-Data-Code-2004-002.pdf>.
- [28] E. Sh. Soukhovitskiĭ, S. Chiba, O. Iwamoto, K. Shibata, T. Fukahori, and G. B. Morogovskii, Programs OPTMAN and SHEMMAN Version 8, Japan Atomic Energy Research Institute Technical Report No. JAERI-Data/Code 2005-002, 2005, <https://jopss.jaea.go.jp/pdfdata/JAERI-Data-Code-2005-002.pdf>.
- [29] E. Sh. Soukhovitskiĭ, S. Chiba, R. Capote, J. M. Quesada, S. Kunieda and G. B. Morogovskii, Japan Atomic Energy Research Institute, Technical Report No. JAERI-Data/Code 2008-025, 2008, <https://jopss.jaea.go.jp/pdfdata/JAEA-Data-Code-2008-025.pdf>.
- [30] E. Sh. Sukhovitskiĭ, Y.-O. Lee, J. Chang, S. Chiba, and O. Iwamoto, Nucleon interaction with ^{58}Ni up to 150 MeV studied in the coupled-channels approach based on the soft-rotator nuclear structure model, *Phys. Rev. C* **62**, 044605 (2000).
- [31] EXchange FORmat database (EXFOR) is maintained by the Network of Nuclear Reaction Data Centers (see www-nds.iaea.org/nrdc/). Data are available online (e.g., at www-nds.iaea.org/exfor/).
- [32] J. P. Delaroche, Y. Wang, and J. Rapaport, Neutron- ^{90}Zr mean field from a dispersive optical model analysis, *Phys. Rev. C* **39**, 391 (1989).